lacking vessels in the xylem. Example: *Hydrilla, Winteraceae* family members.

II) Scattered Vascular bundles along with cambial activity in dicots.

In dicot stem, normally the vascular bundles are arranged in a ring, whereas some dicot stems possess scattered vascular bundles in the cortex like monocot stems. Example: *Piper, Peperomia.*

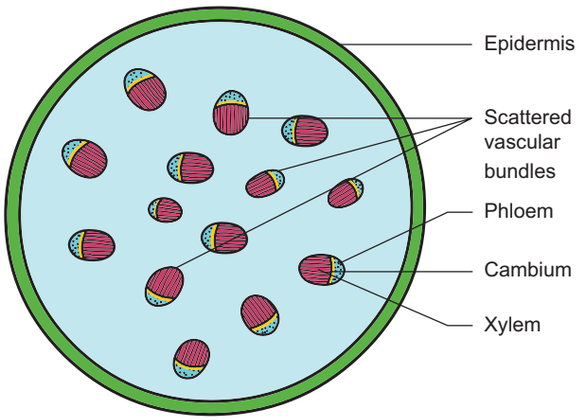


Figure 10.24: Transverse section (T.S) of *Peperomia* stem (diagrammatic) showing anomalous primary growth

10.4 Timber

Timber is derived from wood logs through the process of sawing. There are many methods of sawing. Among them, rib sawing is the most common method in making timber. Timber is mainly used for carpentry and building houses. In order to enrich the quality of timber, seasoning of wood is done. Timber is the most important tissue that sequestrates atmospheric carbon and this reduces global warming.

10.4.1 Seasoning of wood

It is the process in which moisture content from the wood is removed. There are two types of seasoning.

1. Air Seasoning is the process in which the moisture can be removed without resorting to artificial heat. In this method of artificial seasoning, the cut timber pieces are left exposed in the open air and the moisture is removed naturally and gradually by the heat of the sun. It increases the strength, the combustibility and renders the wood less subject to decay.

2. Kiln Seasoning is the process in which the moisture can be removed by artificial method in an enclosed condition. The timber pieces are enclosed in a steam-heater chamber into which air is introduced and circulated by fans, ensuring the removal of moisture uniformly, rapidly and completely.

Activity

List out the uses of saw dust, shavings and wood flour.

10.4.2 Grain, Texture, and Figure of Wood

Grain refers to the structural arrangement of wood, while **texture** shows the relative size and quality of the various types of wood. **Figure of wood** refers to the pattern formed by grains in wood when the wood is cut in the longitudinal direction. It depends on the grain and texture and their exposure by direction of sawing.



Ply wood:

It is manufactured by gluing together 3 to 9 thin layers or piles of wood veneers. It is used in flooring, walls, false roof and vehicle interiors.

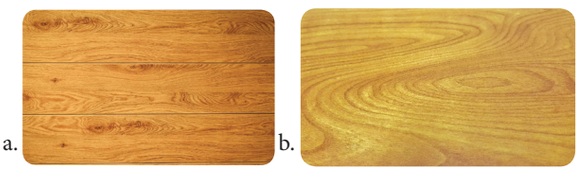


Figure 10.25: Shows grain, texure and figure of wood (a–b).

Activity

Collect some pieces of plywood, analyse the layers and discuss yourself how it is made.

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